

Apple Silicon VMware, ROS2 and Gazebo Installation

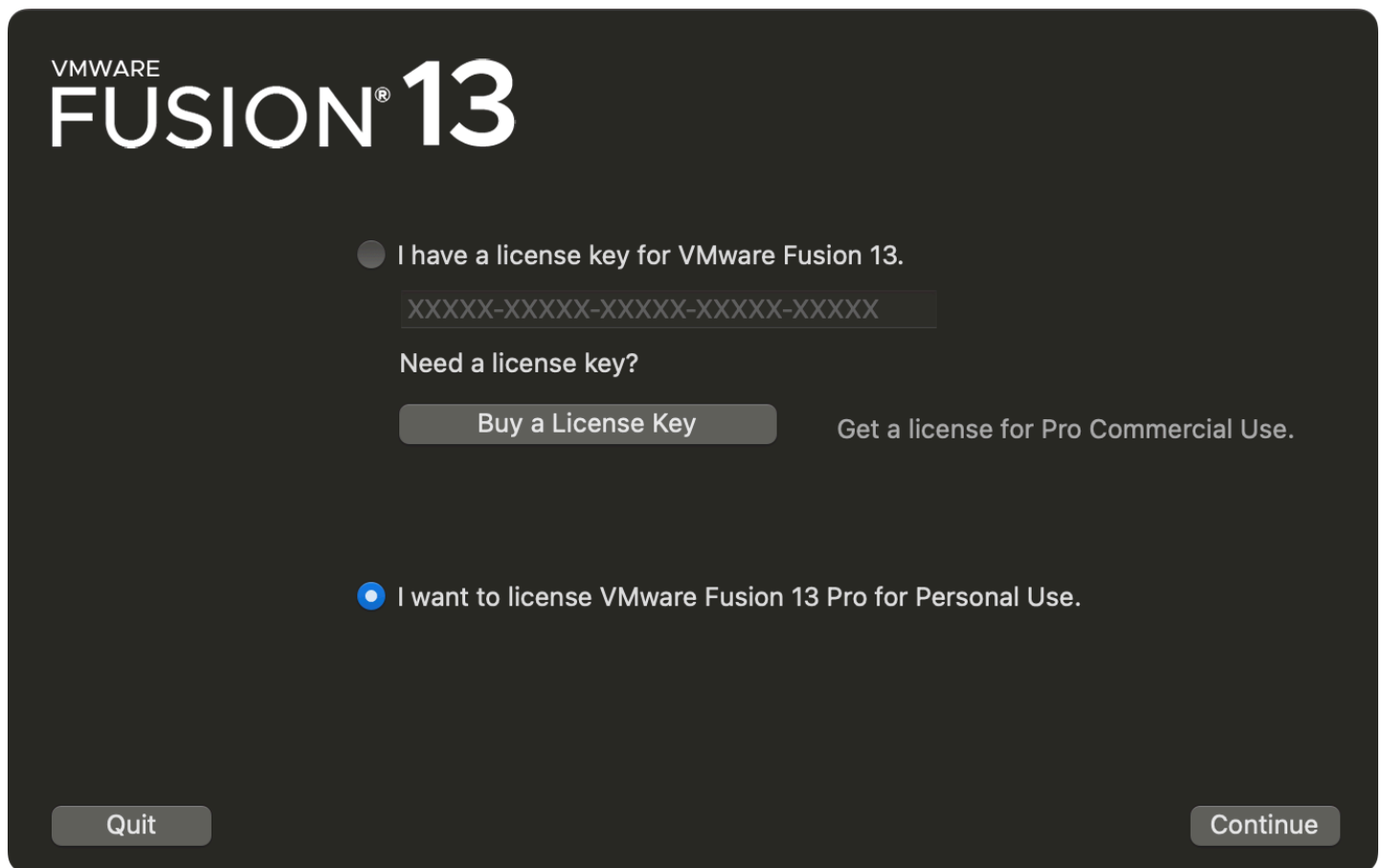
Course: MTE544 via Professor Yue Hu

TA: Minghao Ning

To make it easier for you, we have uploaded the VMware fusion and ubuntu22 iso files

Install VMware Fusion

[VMware-Fusion-13.6.0-24238079_universal.dmg](#)



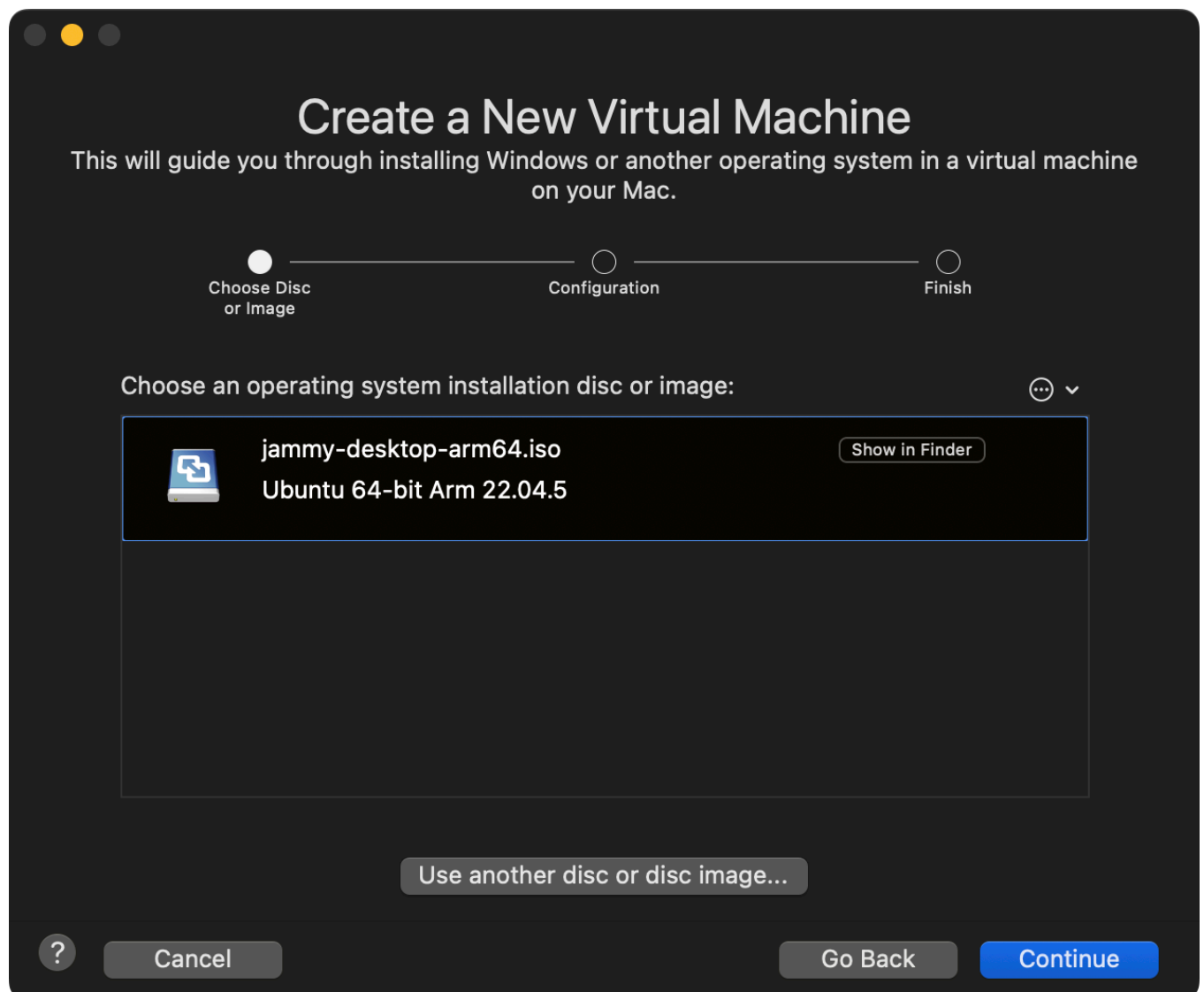
Registration: <https://profile.broadcom.com/web/registration>

Software Download: <https://support.broadcom.com/group/ecx/productfiles?subFamily=VMware%20Fusion&displayGroup=VMware%20Fusion%2013%20Pro%20for%20Personal%20Use&release=13.6&os=&servicePk=522387&language=EN>

Install Ubuntu22 ARM64

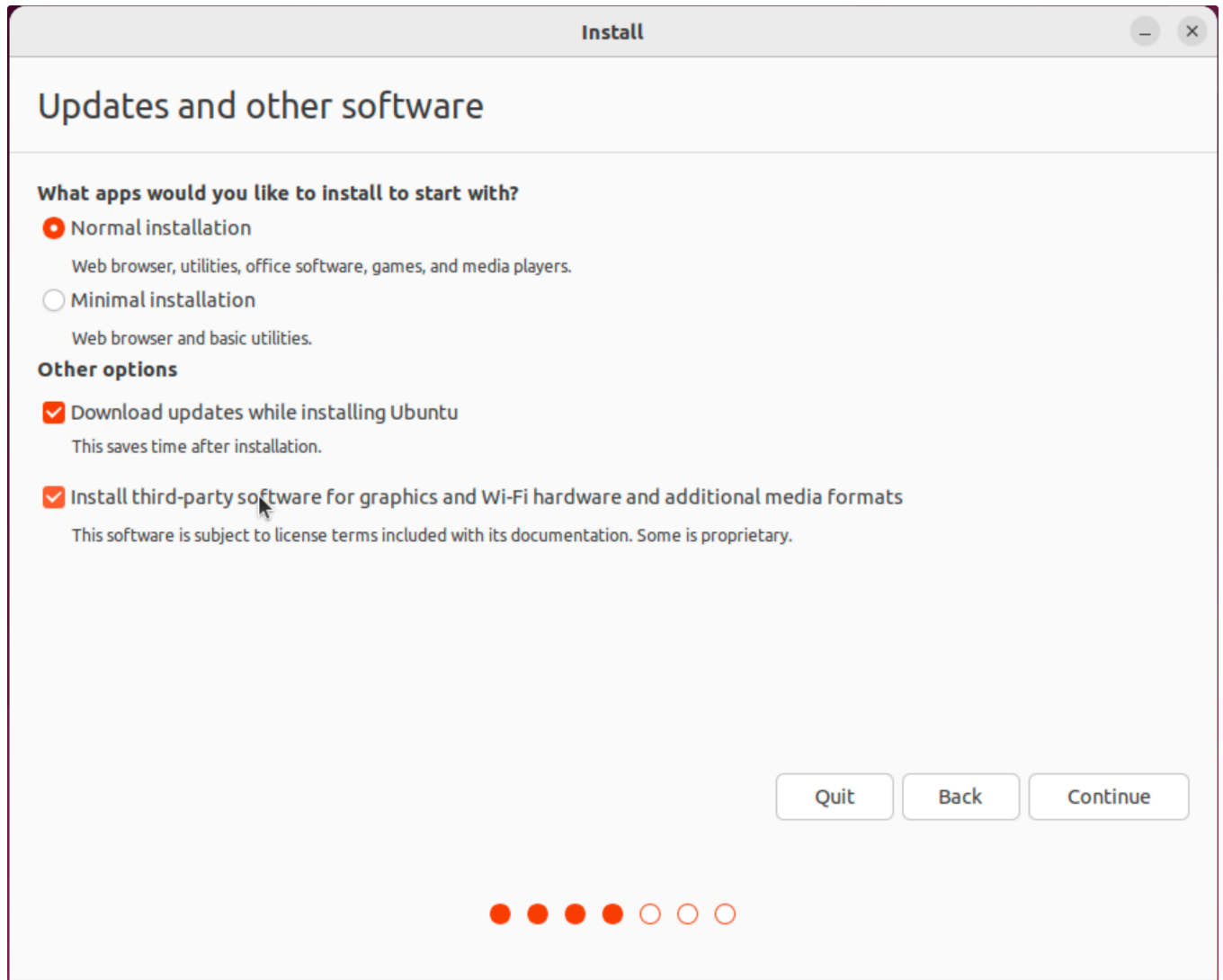
Download the ubuntu22 iso image: <https://cdimage.ubuntu.com/jammy/daily-live/current/jammy-desktop-arm64.iso>

- Install from disc or image
- Drag the [jammy-desktop-arm64.iso](https://cdimage.ubuntu.com/jammy/daily-live/current/jammy-desktop-arm64.iso) into the VMware
- Continue



- Click "Finish"

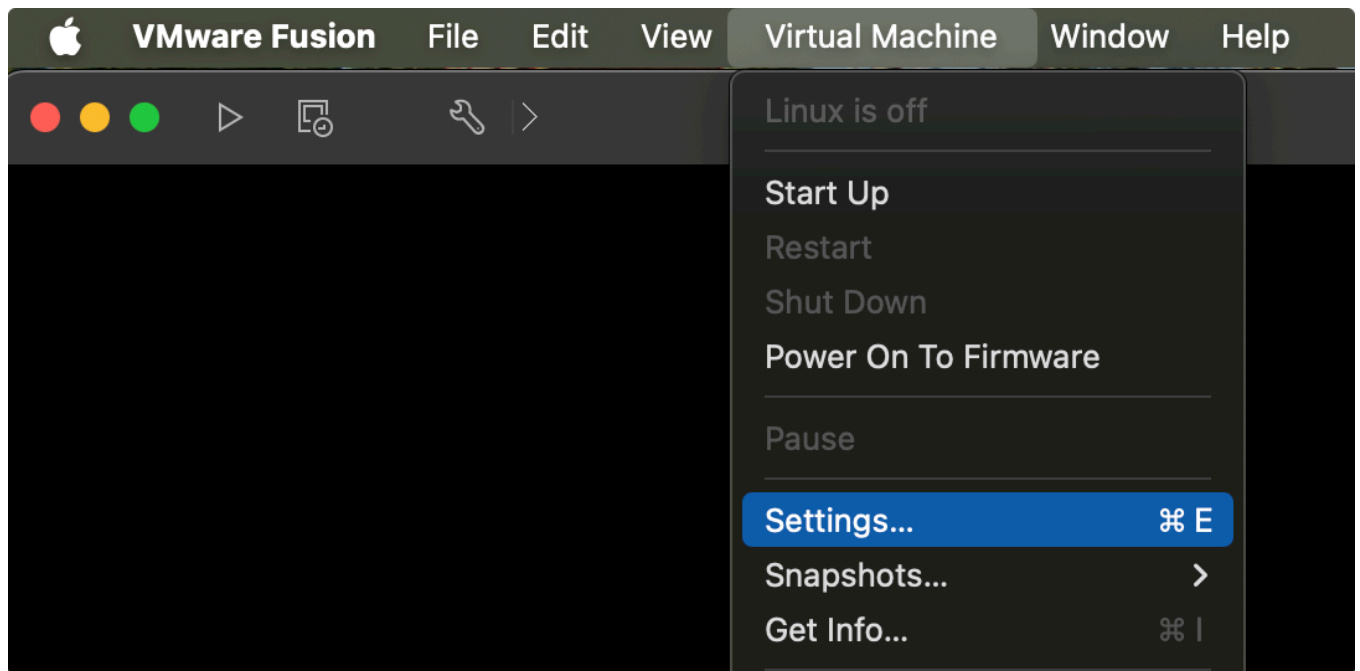
- Then "Try or Install Ubuntu"
- Double click "Install Ubuntu 22.04.5 LTS"
-



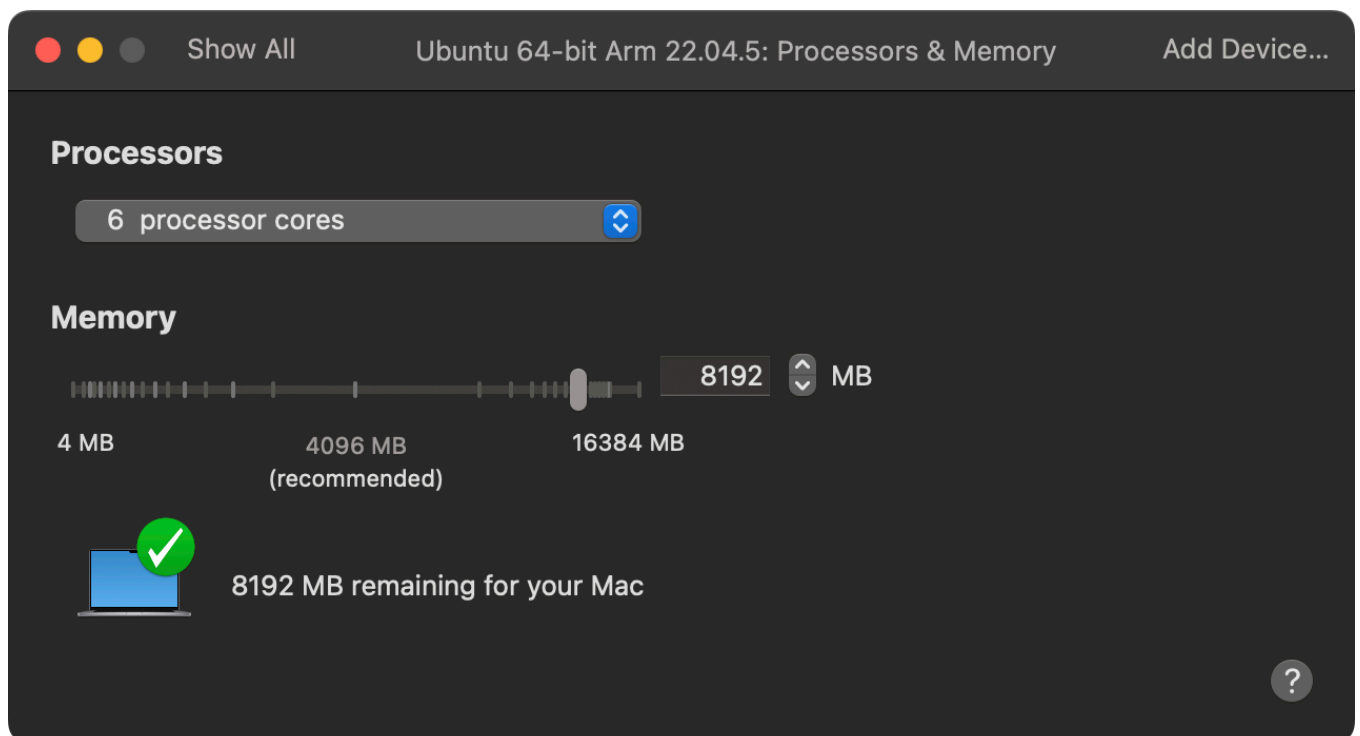
- Click "Erase Disk and Install"
- After the installation, click "Restart Now" and then press "Enter" when it shows "remove installation medium".

Config the Ubuntu22

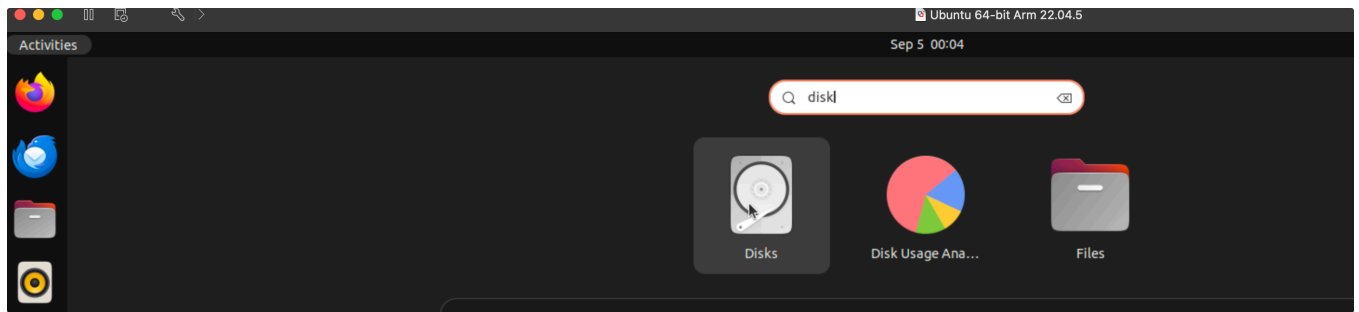
- Open a terminal
- Run `sudo apt-get install open-vm-tools-desktop` in order to share the clipboard between the virtual machine and the Mac computer
- Then power off the ubuntu22



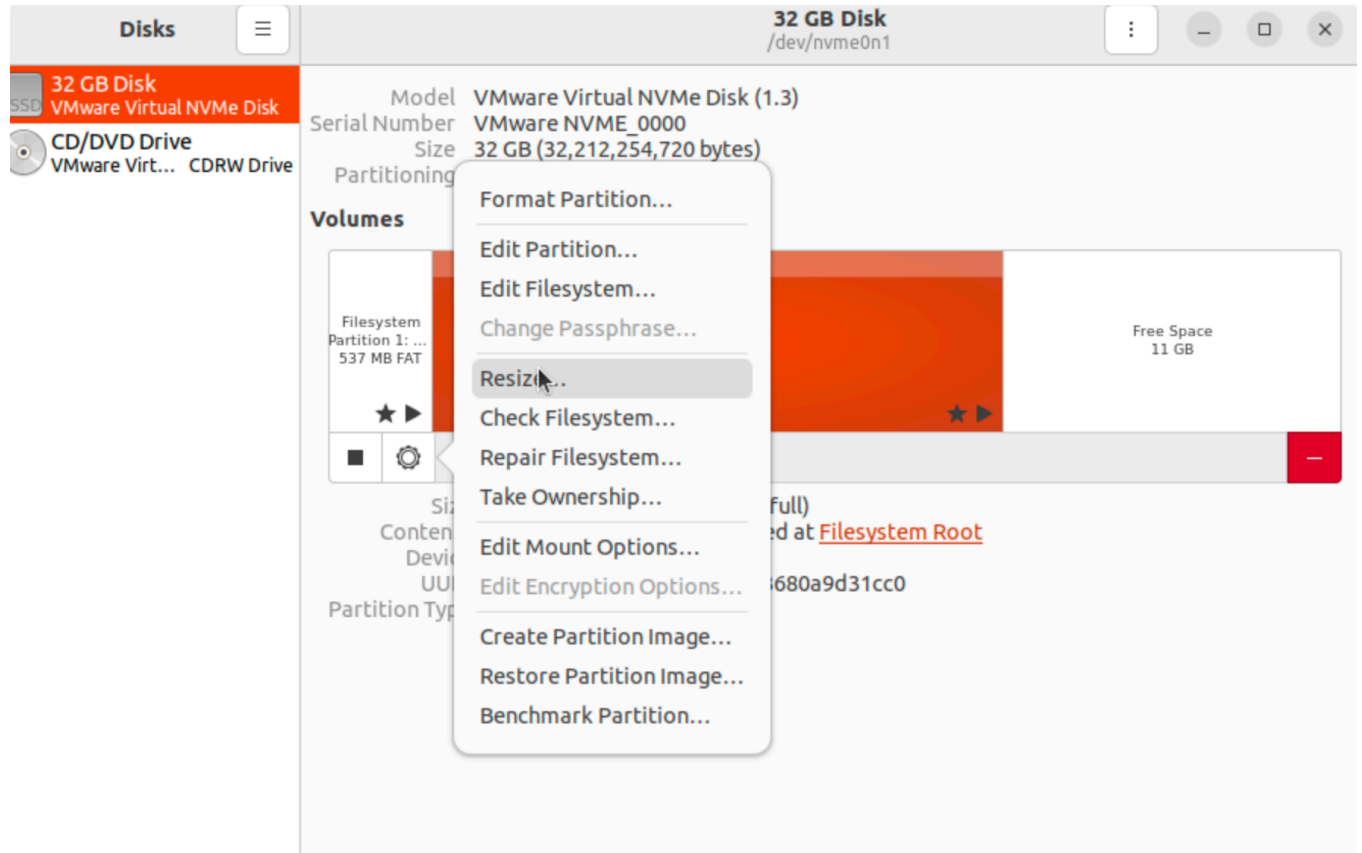
- Go to "Processors & Memory", increase the number of processors and memory

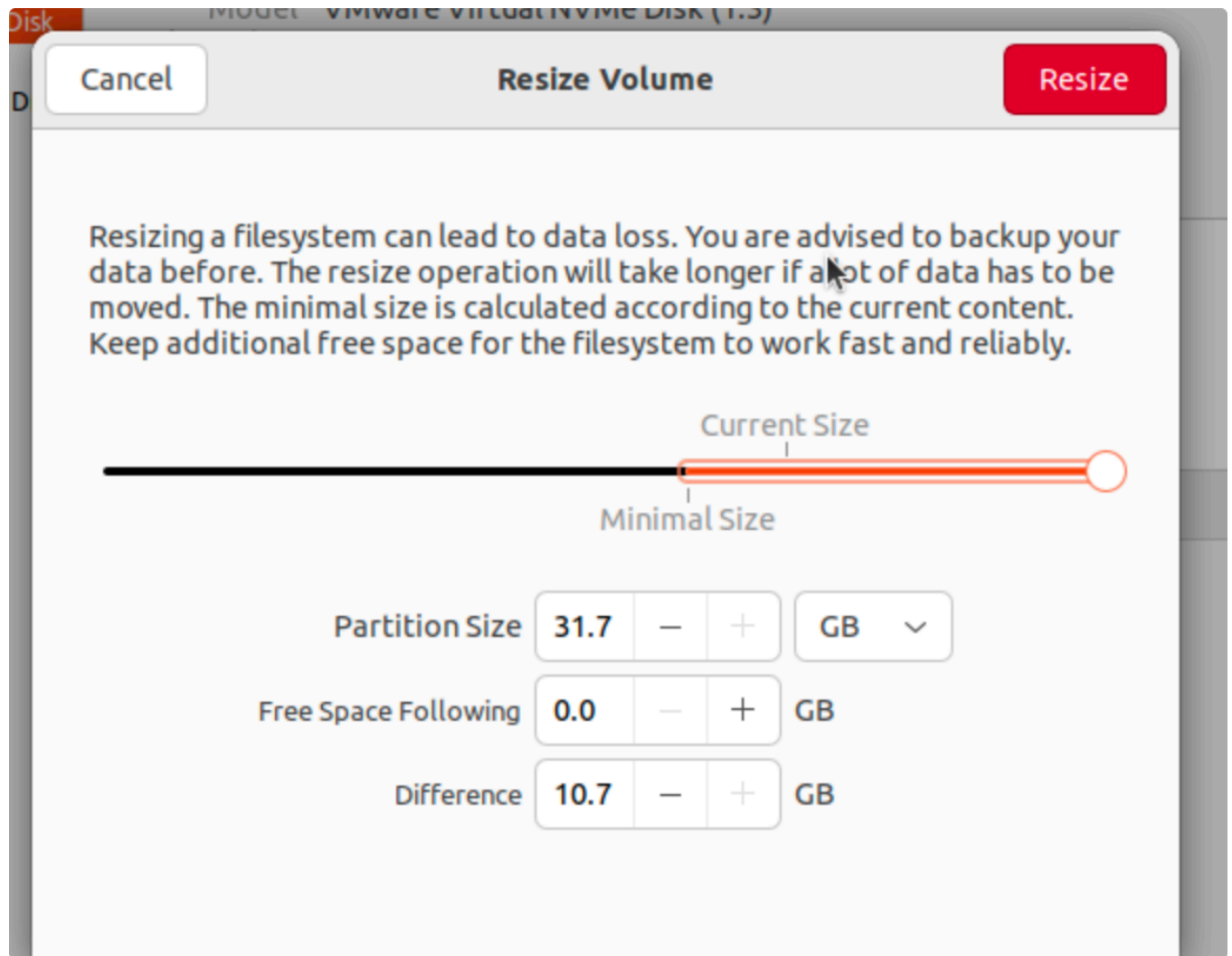


- Go to "Display" and make sure "Accelerate 3D graphics" is **not enabled**.
- Increase the disk space, click "Hard Disk NVMe", increase it to 30GB
- Then turn on the ubuntu22
- Open "Disk"



- Resize the Partition ending with Ext4





Setup the ROS2

Please follow the repo https://github.com/UW-MTE544/MTE544_student/tree/setup to install ROS2. You may need to run `sudo apt install git` if you don't have git

Setup Gazebo

As apple silicon use ARM64 architecture, and the simulation platform gazebo doesn't directly support ARM64, so you have to manually install the [gazebo](#)

1. Install some dependencies

```

sudo apt-add-repository ppa:dartsim
sudo apt update
sudo apt install libdart-dev libdart-utils-dev libdart-external-ikfast-dev
libsdformat9-dev libfreeimage-dev libprotoc-dev libprotobuf-dev protobuf-compiler
freeglut3-dev libcurl4-openssl-dev libtinyxml-dev libtinyxml2-dev libtar-dev libtbb-dev
libogre-1.9-dev libxml2-dev pkg-config qtbase5-dev libqwt-qt5-dev libltdl-dev libgts-
dev libboost-thread-dev libboost-system-dev libboost-filesystem-dev libboost-program-
options-dev libboost-regex-dev libboost-iostreams-dev libsimbody-dev libignition-
common3-dev libignition-fuel-tools4-dev libignition-transport8-dev libignition-math6-
dev libignition-msgs5-dev

```

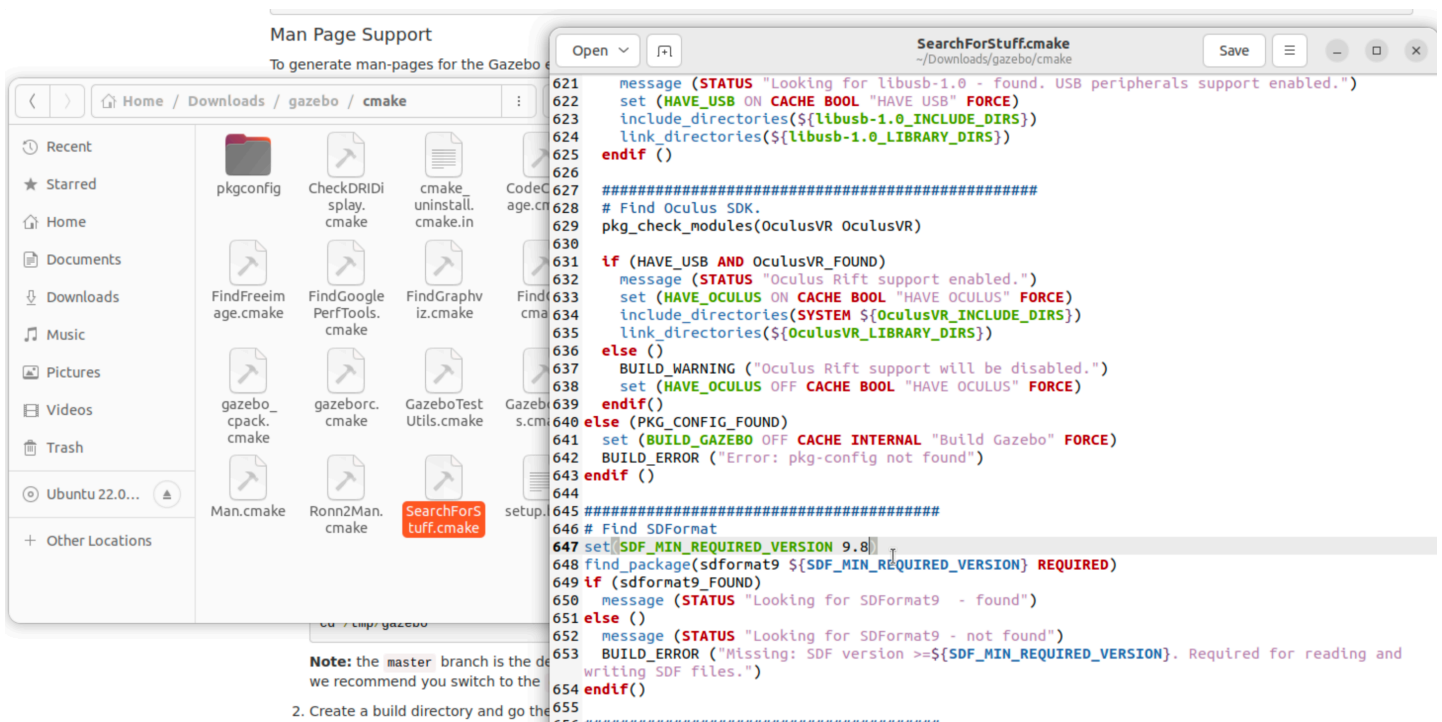
2. Download (or git clone) the gazebo source code

```

# First clone the gazebo git repo
cd ~/Downloads/
git clone https://github.com/osrf/gazebo

```

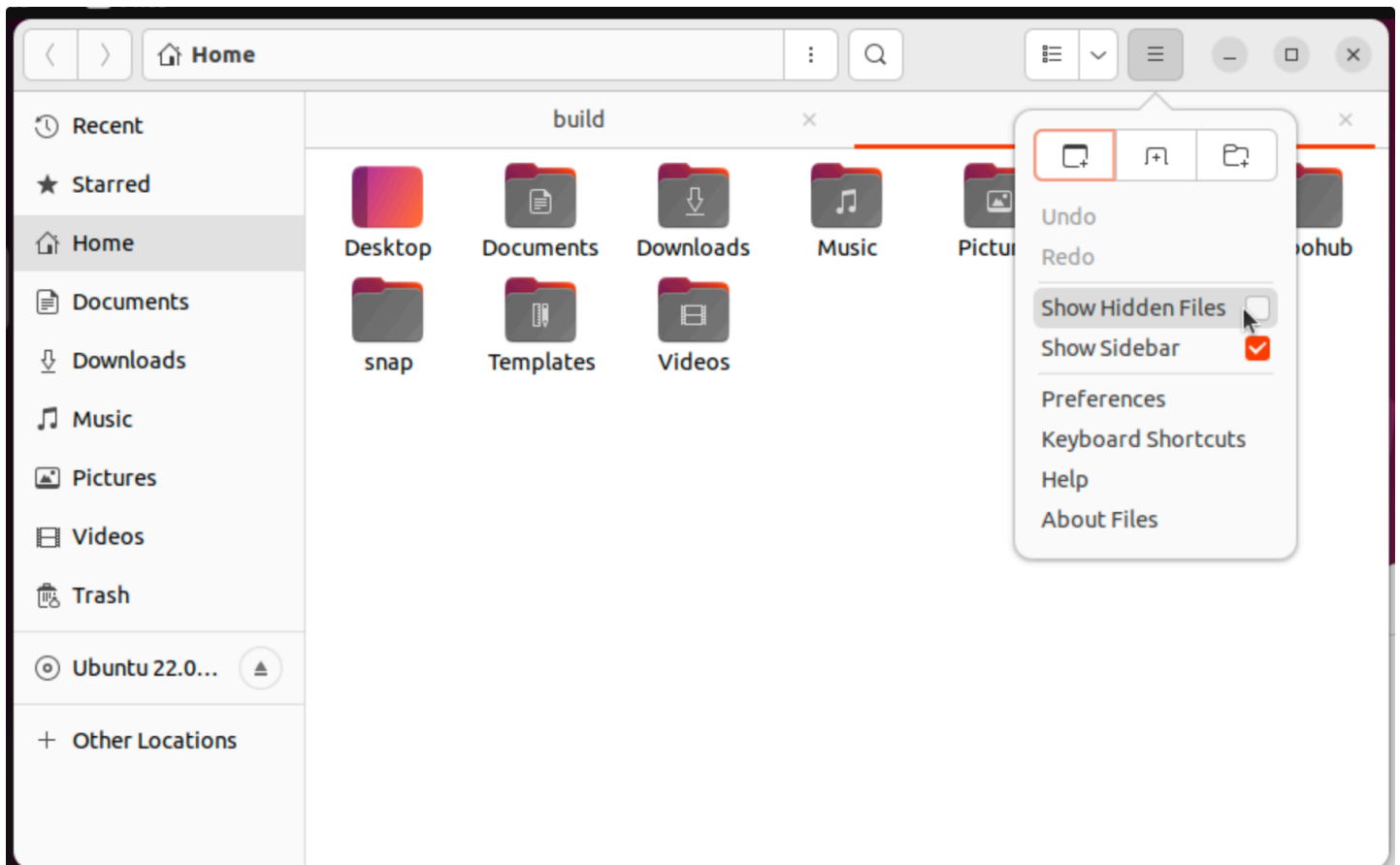
3. Modify the line 647 of SearchForStuff.cmake in Downloads/gazebo/cmake. Change from 9.8 to 9.7 as the default libsdformat version of ubuntu22 is 9.7.



4. Compile the gazebo and install it

```
cd ~/Downloads/gazebo
mkdir build
cd build
cmake ../
make -j3
sudo make install
```

5. Modify the .bashrc, show the hidden file via



6. Add the following lines to the end of .bashrc

```
export LD_LIBRARY_PATH=/usr/local/lib:$LD_LIBRARY_PATH
export PATH=/usr/local/bin:$PATH
export PKG_CONFIG_PATH=/usr/local/lib/pkgconfig:$PKG_CONFIG_PATH
```

```
116 fi
117 fi
118 source /opt/ros/humble/setup.bash
119 export TURTLEBOT3_MODEL=burger
120 export LD_LIBRARY_PATH=/usr/local/lib:$LD_LIBRARY_PATH
121 export PATH=/usr/local/bin:$PATH
122 export PKG_CONFIG_PATH=/usr/local/lib/pkgconfig:$PKG_CONFIG_PATH
```

sh ▾ Tab Width: 8 ▾

Compile the Turtlebot3 Packages

1. Create workspace for compiling

```
cd ~/Downloads
mkdir turtlebot_gazebo_ws && cd turtlebot_gazebo_ws
mkdir src && cd src
```

2. Clone the turtlebot3_simulations

```
cd ~/Downloads/turtlebot_gazebo_ws/src
git clone https://github.com/ROBOTIS-GIT/turtlebot3_simulations
cd turtlebot3_simulations
git checkout ros2
```

3. Clone the gazebo_ros_pkgs

```
cd ~/Downloads/turtlebot_gazebo_ws/src
git clone https://github.com/ros-simulation/gazebo_ros_pkgs
cd gazebo_ros_pkgs
git checkout ros2
```

4. Clone the turtlebot4_pkgs

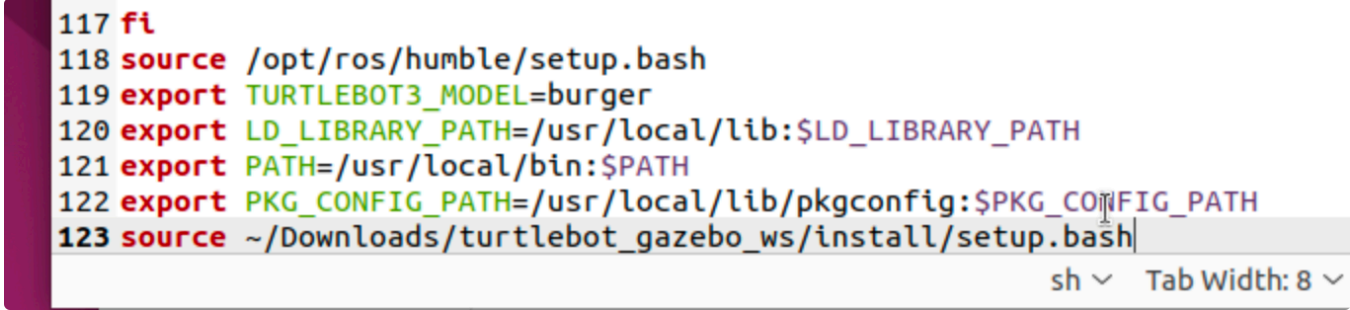
```
cd ~/Downloads/turtlebot_gazebo_ws/src
git clone https://github.com/turtlebot/turtlebot4
cd turtlebot4
git checkout humble
```

4. Compile

```
sudo apt-get install ros-humble-camera-info-manager ros-humble-irobot-create-msgs
cd ~/Downloads/turtlebot_gazebo_ws
export MAKEFLAGS="-j 3"
colcon build
```

5. Add the following line to the end of .bashrc file

```
source ~/Downloads/turtlebot_gazebo_ws/install/setup.bash
```



```
117 fl
118 source /opt/ros/humble/setup.bash
119 export TURTLEBOT3_MODEL=burger
120 export LD_LIBRARY_PATH=/usr/local/lib:$LD_LIBRARY_PATH
121 export PATH=/usr/local/bin:$PATH
122 export PKG_CONFIG_PATH=/usr/local/lib/pkgconfig:$PKG_CONFIG_PATH
123 source ~/Downloads/turtlebot_gazebo_ws/install/setup.bash
```

sh ▾ Tab Width: 8 ▾

Test the installation

```
ros2 launch turtlebot3_gazebo turtlebot3_house.launch.py
ros2 launch turtlebot4_navigation slam.launch.py
ros2 launch turtlebot4_viz view_robot.launch.py
ros2 run teleop_twist_keyboard teleop_twist_keyboard
```